Cart-Pole Inverted Pendulum – Modelling and PID Control

Project Overview

This project focuses on the mathematical modelling, dynamic simulation and PID-based control of a cart-pole inverted pendulum system.

The system consists of a pendulum mounted on a horizontally moving cart. The main objective is to maintain the pendulum in its unstable upright position while controlling the horizontal movement of the cart.

The project was developed using MATLAB Simulink for system modelling and control simulation, with SolidWorks used for mechanical CAD design.

Project Objectives

- Develop a dynamic model of the cart-pole inverted pendulum.
- Analyse the nonlinear behaviour of the system.
- Design separate PID controllers for cart position and pendulum angle.
- Implement the control architecture in MATLAB Simulink.
- Evaluate system performance through simulation.
- Develop a 3D mechanical design using SolidWorks.

Software and Tools

- MATLAB Simulink
- SolidWorks
- PID Control
- Dynamic System Modelling
- Simulation and Response Analysis

Project Components

1. Simulink Model

Dynamic system modelling and implementation of the PID control architecture.

2. Mechanical CAD Design

3D modelling and mechanical assembly using SolidWorks.

3. Simulation and Results

Analysis of pendulum angular response, cart position response and system movement.